

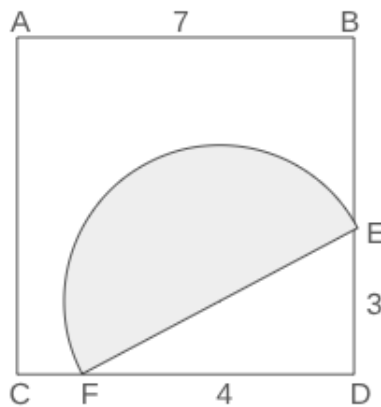
Pre-Algebra Written Exam

2026 James Clemens Math Tournament

1. You will have 90 minutes to complete this examination.
2. This exam consists of 25 multiple choice questions and 3 tiebreaker questions. The multiple choice questions are each worth 4 points if answered correctly and -1 points if answered incorrectly. There is no penalty for unanswered questions. The tiebreaker questions are worth 0.1 points if answered correctly and 0 points if answered incorrectly or left unanswered.
3. Use a #2 pencil.
4. Calculators, cellular devices, books, notes, and other aids are prohibited during this exam. Scratch paper will be provided for calculations. If you require additional scratch paper, simply raise your hand and your proctor will assist you.
5. Mark your answers to the questions in the Scantron form provided. You may use the test booklets for scratch work, but only the answers marked in the Scantron form will be counted. Make sure to fill in the bubbles on the Scantron form **completely** and erase any stray marks.
6. Write your answers to the tiebreaker questions on the back of the Scantron form. Label the tiebreakers as TB1, TB2, and TB3.
7. NOTA denotes "None Of The Above."
8. Diagrams are not necessarily drawn to scale.
9. In the event of a tie, answers will be evaluated backwards starting from multiple choice question 25 to determine a winner.
10. Although this math tournament is intended to demonstrate your knowledge and skills in math, be sure to enjoy the experience and have fun!
11. You may keep the test booklet after the exam.

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- A line with equation $y = 2x + 5$ is parallel to a line that contains the point $(1, 3)$. What is the equation of the line?
 (A) $y = \frac{1}{2}x + 1$ (B) $y = 3x + 2$ (C) $y = -\frac{1}{2}x + \frac{7}{2}$ (D) $y = 2x + 1$
 (E) NOTA
- How many ways are there to arrange the letters in “James Clemens” if the letters in “James” must appear together and in the front, and those in “Clemens” must appear together and in the back? For example, “JMASE CELMESN” works, but “JAME CSLMESNE” or “CELMESN JMASE” does not.
 (A) 162,400 (B) 372 (C) 324,800 (D) 302,400 (E) NOTA
- A meteorologist is tracking a tornado. He needs to read the clock to determine the current time, but the power has gone out and it is too dark to read the numbers. However, he can see that the minute hand is at 3 and the angle between the minute and hour hand is 67.5 degrees. What time is it?
 (A) 7 : 30 (B) 7 : 15 (C) 5 : 15 (D) 2 : 15 (E) NOTA
- Cathy is making sugar cookies. She needs her icing to be 10% sugar and have a total of one cup of icing. If she made two previous attempts she wants to mix together to make her final icing mixture: the first attempt is 50% sugar, and the second attempt is 5% sugar, how many cups of the first mixture should she use to create the final mixture?
 (A) $\frac{8}{9}$ (B) $\frac{1}{9}$ (C) $\frac{1}{2}$ (D) $\frac{1}{5}$ (E) NOTA
- Solve for the area of the shaded region if quadrilateral $ABCD$ is a square, FD has length 4, ED has length 3, and AB has length 7?



- (A) 3.125π (B) 6.25π (C) 12.5π (D) 25π (E) NOTA

6. Solve for x :

$$\sqrt{x+5} = \sqrt{2x+1}$$

- (A) 9 (B) $\sqrt{3}$ (C) 6 (D) -4 (E) NOTA

7. Hailey and Alice are completing math problems together on a ciphering test. Hailey solves five problems in four minutes while Alice solves ten problems in nine minutes. How long, in minutes, will it take them to solve 85 questions?

- (A) 50 (B) 65 (C) 36 (D) 5.7 (E) NOTA

8. George is making ice cream cones. He fills the entire cone without going over the rim of the cone. He fills each cone from a cylindrical tub that has the same height and radius as the cone. How many ice cream cones can he make with one tub of ice cream?

- (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) 2 (D) 3 (E) NOTA

9. What is the area of a circle with its center at $(1, 3)$ and passes through the point $(5, 12)$?

- (A) 97π (B) 261π (C) 16π (D) 81π (E) NOTA

10. Kirk and William are playing a game involving flipping two coins. The first coin is fair, while the second has a two times greater chance of landing on heads than tails. Both coins are flipped on a single turn. The first person to get two tails on the same turn wins. If Kirk goes first, what is the probability he wins on his first turn?

- (A) $\frac{1}{3}$ (B) $\frac{1}{6}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$ (E) NOTA

11. In a teacher's class, there are six students. On the most recent test, her class average excluding the lowest grade is 87.4, and the class average excluding the highest grade is 76. What is the difference between the highest and lowest test scores?

- (A) 5.7 (B) 57 (C) 11.4 (D) 28.5 (E) NOTA

12. Simplify:

$$\frac{x^2 - 4}{x^2 + x - 6} \div \frac{x + 2}{x^2 - 3x - 18}$$

- (A) $x - 6$ (B) $\frac{x^2 + 4x + 4}{x^3 - 27x - 54}$ (C) $\frac{2x^2 - 6x - 9}{x + 3}$ (D) $-\frac{1}{4}$
(E) NOTA

13. The operation $x\#y$ is defined as $\frac{x+y}{y}$. For what values of x and y is $x\#y$ defined?

- (A) All values of x and $y \neq 0$ (B) All values of x and all values of y
(C) $x \neq 0$ and $y \neq 0$ (D) $x \neq 10$ and all values of y
(E) NOTA

14. The functions $g(x)$, $f(x)$, and $h(x)$ are defined below. What is $g(f(h(10)))$?

$$g(x) = |x - 300| + 5$$

$$f(x) = x^2 + \sqrt{x}$$

$$h(x) = 2x - 4$$

- (A) -501600 (B) -159 (C) 10 (D) 45 (E) NOTA

15. An arithmetic sequence has first term 3 and common difference 12. What is the sum of the first twenty terms of this sequence?

- (A) 723 (B) 2340 (C) 231 (D) 2316 (E) NOTA

16. Given the system below, what is $x + y + z$?

$$x + 3y + 4z = 19$$

$$2x + 5y = 4z$$

$$3x + 16y = 35$$

- (A) -35.4 (B) 11 (C) 0 (D) 6 (E) NOTA

17. Let $C_1, C_2, \dots, C_{20}$ be a series of 20 concentric circles, where circle C_k has radius k . Sumehra colors circle C_1 blue, the area between C_2 and C_3 blue, the area between C_4 and C_5 blue, and so on. What is the total area of blue in her drawing?

- (A) 210π (B) 1540π (C) 1330π (D) 190π (E) NOTA

18. Lilly writes out all the numbers from 1 to 100, inclusive. How many times does she write the digit 9?

- (A) 10 (B) 19 (C) 20 (D) 11 (E) NOTA

19. Simplify:

$$\frac{(a^5)^2(b^4)^{\frac{1}{2}}}{a^{12} \cdot b \cdot b}$$

- (A) $\frac{b}{a^2}$ (B) $\frac{b^{\frac{5}{2}}}{a^5}$ (C) $a^{\frac{5}{6}}$ (D) a^2 (E) NOTA

20. The last two digits of an integer x are 24. Which of the following numbers is a divisor of x , regardless of the value of the remaining digits of x ?

- (A) 3 (B) 12 (C) 4 (D) 6 (E) NOTA

21. What is the sum of the interior angles of a regular decagon?

- (A) 900 (B) 1440 (C) 1800 (D) 144 (E) NOTA

22. How many positive factors does 6000 have?

- (A) 3 (B) 40 (C) 11 (D) 12 (E) NOTA

23. A student writes the first 100 triangular numbers. A triangular number is a number that can be written as $1 + 2 + \cdots + (n - 1) + n$ for some positive integer n . If he crosses out all those divisible by at least one odd number, how many numbers will be left?

- (A) 0 (B) 50 (C) 66 (D) 100 (E) NOTA

24. What is the remainder when 33^{2026} is divided by 10?

- (A) 1 (B) 3 (C) 7 (D) 9 (E) NOTA

25. Solve for all values of x :

$$x^2 + 8x + 15 = 0$$

- (A) 3 and 5 (B) $\frac{5}{3}$ (C) -3 and -5 (D) $-\frac{5}{3}$
(E) NOTA

Tiebreakers:

TB1. What is the sum of the solutions of $|2x + 4| = 16$?

TB2. A triangle has side lengths 2 and 5. How many possible integer values are there for the third side?

TB3. What is the area of an equilateral triangle with side length 4? Express your answer in simplest radical form.